

Assignment_1

February 28, 2025

```
[ ]: Name : Omkar Hulawale  
Class , Batch : TE-A , A3 Batch  
Roll No: 13165
```

```
[3]: import pandas as pd  
import numpy as np
```

```
[5]: df=pd.read_csv(r"C:\Users\hulaw\OneDrive\Desktop\DSBDA\data.csv")  
df
```

```
[5]:
```

	Unnamed: 0	A	B	C	D
0	0	-1.557181	0.373416	1.040685	-0.648513
1	1	-0.835872	-0.005259	0.502099	-0.956839
2	2	-0.257153	2.638228	-0.535731	-1.069443
3	3	-0.505502	-0.339260	0.602622	-1.420995
4	4	1.366398	-0.368296	0.642091	1.609680
5	5	-1.350101	0.210097	-0.743280	0.035729
6	6	-0.621305	-0.620733	0.141429	-1.831875
7	7	-0.670370	-0.379230	1.950228	-0.487202
8	8	-1.360577	0.044675	-0.576289	0.998686
9	9	-0.120302	-0.146491	-0.420956	1.412786
10	10	0.216936	0.398267	0.342519	0.387263
11	11	0.567832	-2.023668	-0.315318	-0.064349
12	12	-0.161148	-0.118924	-1.226679	-0.932540
13	13	-1.063620	0.041304	-1.037695	0.628520
14	14	0.912931	0.848745	-1.215276	-0.024202
15	15	-0.661489	-1.543994	-0.452754	-1.589521
16	16	-0.272063	-1.012084	0.414147	-0.021171
17	17	0.468146	0.232378	-1.890289	0.827622
18	18	1.744694	0.052282	0.890462	-0.155540
19	19	0.381029	0.666666	0.957832	1.476093
20	20	0.503970	0.567258	1.578919	1.041839
21	21	-0.561937	0.534243	0.246232	0.763021
22	22	0.064446	-0.483406	-0.584869	-1.066767
23	23	2.762885	0.060047	-0.709070	-1.350154
24	24	-0.399912	0.755714	-0.138098	0.771859
25	25	2.289132	-0.178023	0.392030	-1.638884
26	26	-1.667799	-0.723245	-0.298492	-0.464406

27	27	0.871473	-0.875910	1.693428	-0.120278
28	28	2.035892	-0.442804	1.710999	-0.164896
29	29	2.646331	0.006441	-0.819143	-0.140657
30	30	-1.645022	1.841754	-0.390258	-0.118842
31	31	-2.085251	0.043444	-0.183578	0.548298
32	32	1.427635	-0.349981	1.038814	-0.955178
33	33	-0.680210	0.566180	-0.325638	0.244383
34	34	-0.026502	0.758021	0.466427	-0.198146
35	35	0.433309	0.623019	-0.201789	1.983037
36	36	0.646673	-0.715421	-0.322057	0.645001
37	37	0.048681	-1.203145	0.460026	0.192644
38	38	0.098718	-0.672915	-0.229832	-0.187133
39	39	-0.059930	-1.472300	-1.281250	-0.666214
40	40	1.515724	0.207993	-1.037067	-0.393363
41	41	-0.949868	-1.115604	-0.554789	1.324991
42	42	-1.452316	-0.572414	-0.808790	1.615190
43	43	-1.117542	0.083952	-0.874761	0.884389
44	44	0.834118	0.423509	-0.565575	-0.169843
45	45	-0.382312	0.175994	-0.820539	-0.542783
46	46	-0.982393	-0.396464	0.550612	1.584936
47	47	0.581846	-0.994098	0.519327	1.286765
48	48	-0.412282	-1.889874	-0.047348	-0.767669
49	49	0.446292	0.326242	0.230500	0.841655

```
[7]: df.isnull()
```

```
[7]:
```

	Unnamed: 0	A	B	C	D
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
5	False	False	False	False	False
6	False	False	False	False	False
7	False	False	False	False	False
8	False	False	False	False	False
9	False	False	False	False	False
10	False	False	False	False	False
11	False	False	False	False	False
12	False	False	False	False	False
13	False	False	False	False	False
14	False	False	False	False	False
15	False	False	False	False	False
16	False	False	False	False	False
17	False	False	False	False	False
18	False	False	False	False	False
19	False	False	False	False	False

20	False	False	False	False	False
21	False	False	False	False	False
22	False	False	False	False	False
23	False	False	False	False	False
24	False	False	False	False	False
25	False	False	False	False	False
26	False	False	False	False	False
27	False	False	False	False	False
28	False	False	False	False	False
29	False	False	False	False	False
30	False	False	False	False	False
31	False	False	False	False	False
32	False	False	False	False	False
33	False	False	False	False	False
34	False	False	False	False	False
35	False	False	False	False	False
36	False	False	False	False	False
37	False	False	False	False	False
38	False	False	False	False	False
39	False	False	False	False	False
40	False	False	False	False	False
41	False	False	False	False	False
42	False	False	False	False	False
43	False	False	False	False	False
44	False	False	False	False	False
45	False	False	False	False	False
46	False	False	False	False	False
47	False	False	False	False	False
48	False	False	False	False	False
49	False	False	False	False	False

```
[9]: df.notnull()
```

```
[9]:
```

	Unnamed: 0	A	B	C	D
0	True	True	True	True	True
1	True	True	True	True	True
2	True	True	True	True	True
3	True	True	True	True	True
4	True	True	True	True	True
5	True	True	True	True	True
6	True	True	True	True	True
7	True	True	True	True	True
8	True	True	True	True	True
9	True	True	True	True	True
10	True	True	True	True	True
11	True	True	True	True	True
12	True	True	True	True	True

13	True	True	True	True	True
14	True	True	True	True	True
15	True	True	True	True	True
16	True	True	True	True	True
17	True	True	True	True	True
18	True	True	True	True	True
19	True	True	True	True	True
20	True	True	True	True	True
21	True	True	True	True	True
22	True	True	True	True	True
23	True	True	True	True	True
24	True	True	True	True	True
25	True	True	True	True	True
26	True	True	True	True	True
27	True	True	True	True	True
28	True	True	True	True	True
29	True	True	True	True	True
30	True	True	True	True	True
31	True	True	True	True	True
32	True	True	True	True	True
33	True	True	True	True	True
34	True	True	True	True	True
35	True	True	True	True	True
36	True	True	True	True	True
37	True	True	True	True	True
38	True	True	True	True	True
39	True	True	True	True	True
40	True	True	True	True	True
41	True	True	True	True	True
42	True	True	True	True	True
43	True	True	True	True	True
44	True	True	True	True	True
45	True	True	True	True	True
46	True	True	True	True	True
47	True	True	True	True	True
48	True	True	True	True	True
49	True	True	True	True	True

```
[11]: df.isnull().sum()
```

```
[11]: Unnamed: 0    0
      A          0
      B          0
      C          0
      D          0
      dtype: int64
```

```
[13]: df.notnull().sum()
```

```
[13]: Unnamed: 0    50  
      A          50  
      B          50  
      C          50  
      D          50  
      dtype: int64
```

```
[15]: df.describe()
```

```
[15]:
```

	Unnamed: 0	A	B	C	D
count	50.00000	50.000000	50.000000	50.000000	50.000000
mean	24.50000	0.020103	-0.123273	-0.044716	0.059140
std	14.57738	1.139660	0.845836	0.848863	0.966854
min	0.00000	-2.085251	-2.023668	-1.890289	-1.831875
25%	12.25000	-0.677750	-0.608653	-0.582724	-0.622080
50%	24.50000	-0.090116	0.000591	-0.215810	-0.091596
75%	36.75000	0.578342	0.361622	0.493181	0.813681
max	49.00000	2.762885	2.638228	1.950228	1.983037

```
[17]: df.size
```

```
[17]: 250
```

```
[19]: df.ndim
```

```
[19]: 2
```

```
[21]: df.shape
```

```
[21]: (50, 5)
```

```
[23]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 50 entries, 0 to 49  
Data columns (total 5 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   Unnamed: 0   50 non-null     int64  
1   A            50 non-null     float64  
2   B            50 non-null     float64  
3   C            50 non-null     float64  
4   D            50 non-null     float64  
dtypes: float64(4), int64(1)  
memory usage: 2.1 KB
```

```
[25]: df["A"].astype(int)
```

```
[25]: 0    -1  
      1     0  
      2     0  
      3     0  
      4     1  
      5    -1  
      6     0  
      7     0  
      8    -1  
      9     0  
     10     0  
     11     0  
     12     0  
     13    -1  
     14     0  
     15     0  
     16     0  
     17     0  
     18     1  
     19     0  
     20     0  
     21     0  
     22     0  
     23     2  
     24     0  
     25     2  
     26    -1  
     27     0  
     28     2  
     29     2  
     30    -1  
     31    -2  
     32     1  
     33     0  
     34     0  
     35     0  
     36     0  
     37     0  
     38     0  
     39     0  
     40     1  
     41     0  
     42    -1  
     43    -1  
     44     0
```

```
45    0
46    0
47    0
48    0
49    0
Name: A, dtype: int32
```

```
[27]: df.dropna()
```

```
[27]:
```

	Unnamed: 0	A	B	C	D
0	0	-1.557181	0.373416	1.040685	-0.648513
1	1	-0.835872	-0.005259	0.502099	-0.956839
2	2	-0.257153	2.638228	-0.535731	-1.069443
3	3	-0.505502	-0.339260	0.602622	-1.420995
4	4	1.366398	-0.368296	0.642091	1.609680
5	5	-1.350101	0.210097	-0.743280	0.035729
6	6	-0.621305	-0.620733	0.141429	-1.831875
7	7	-0.670370	-0.379230	1.950228	-0.487202
8	8	-1.360577	0.044675	-0.576289	0.998686
9	9	-0.120302	-0.146491	-0.420956	1.412786
10	10	0.216936	0.398267	0.342519	0.387263
11	11	0.567832	-2.023668	-0.315318	-0.064349
12	12	-0.161148	-0.118924	-1.226679	-0.932540
13	13	-1.063620	0.041304	-1.037695	0.628520
14	14	0.912931	0.848745	-1.215276	-0.024202
15	15	-0.661489	-1.543994	-0.452754	-1.589521
16	16	-0.272063	-1.012084	0.414147	-0.021171
17	17	0.468146	0.232378	-1.890289	0.827622
18	18	1.744694	0.052282	0.890462	-0.155540
19	19	0.381029	0.666666	0.957832	1.476093
20	20	0.503970	0.567258	1.578919	1.041839
21	21	-0.561937	0.534243	0.246232	0.763021
22	22	0.064446	-0.483406	-0.584869	-1.066767
23	23	2.762885	0.060047	-0.709070	-1.350154
24	24	-0.399912	0.755714	-0.138098	0.771859
25	25	2.289132	-0.178023	0.392030	-1.638884
26	26	-1.667799	-0.723245	-0.298492	-0.464406
27	27	0.871473	-0.875910	1.693428	-0.120278
28	28	2.035892	-0.442804	1.710999	-0.164896
29	29	2.646331	0.006441	-0.819143	-0.140657
30	30	-1.645022	1.841754	-0.390258	-0.118842
31	31	-2.085251	0.043444	-0.183578	0.548298
32	32	1.427635	-0.349981	1.038814	-0.955178
33	33	-0.680210	0.566180	-0.325638	0.244383
34	34	-0.026502	0.758021	0.466427	-0.198146
35	35	0.433309	0.623019	-0.201789	1.983037
36	36	0.646673	-0.715421	-0.322057	0.645001

37	37	0.048681	-1.203145	0.460026	0.192644
38	38	0.098718	-0.672915	-0.229832	-0.187133
39	39	-0.059930	-1.472300	-1.281250	-0.666214
40	40	1.515724	0.207993	-1.037067	-0.393363
41	41	-0.949868	-1.115604	-0.554789	1.324991
42	42	-1.452316	-0.572414	-0.808790	1.615190
43	43	-1.117542	0.083952	-0.874761	0.884389
44	44	0.834118	0.423509	-0.565575	-0.169843
45	45	-0.382312	0.175994	-0.820539	-0.542783
46	46	-0.982393	-0.396464	0.550612	1.584936
47	47	0.581846	-0.994098	0.519327	1.286765
48	48	-0.412282	-1.889874	-0.047348	-0.767669
49	49	0.446292	0.326242	0.230500	0.841655

```
[29]: df["A"].astype(float)
```

```
[29]: 0    -1.557181
      1    -0.835872
      2    -0.257153
      3    -0.505502
      4     1.366398
      5    -1.350101
      6    -0.621305
      7    -0.670370
      8    -1.360577
      9    -0.120302
     10     0.216936
     11     0.567832
     12    -0.161148
     13    -1.063620
     14     0.912931
     15    -0.661489
     16    -0.272063
     17     0.468146
     18     1.744694
     19     0.381029
     20     0.503970
     21    -0.561937
     22     0.064446
     23     2.762885
     24    -0.399912
     25     2.289132
     26    -1.667799
     27     0.871473
     28     2.035892
     29     2.646331
     30    -1.645022
```



```

31    -2.085251
32     1.427635
33    -0.680210
34    -0.026502
35     0.433309
36     0.646673
37     0.048681
38     0.098718
39    -0.059930
40     1.515724
41    -0.949868
42    -1.452316
43    -1.117542
44     0.834118
45    -0.382312
46    -0.982393
47     0.581846
48    -0.412282
49     0.446292
Name: A, dtype: float64

```

```
[31]: df
```

```

[31]:      Unnamed: 0      A      B      C      D
0         0  -1.557181  0.373416  1.040685 -0.648513
1         1  -0.835872 -0.005259  0.502099 -0.956839
2         2  -0.257153  2.638228 -0.535731 -1.069443
3         3  -0.505502 -0.339260  0.602622 -1.420995
4         4   1.366398 -0.368296  0.642091  1.609680
5         5  -1.350101  0.210097 -0.743280  0.035729
6         6  -0.621305 -0.620733  0.141429 -1.831875
7         7  -0.670370 -0.379230  1.950228 -0.487202
8         8  -1.360577  0.044675 -0.576289  0.998686
9         9  -0.120302 -0.146491 -0.420956  1.412786
10        10   0.216936  0.398267  0.342519  0.387263
11        11   0.567832 -2.023668 -0.315318 -0.064349
12        12  -0.161148 -0.118924 -1.226679 -0.932540
13        13  -1.063620  0.041304 -1.037695  0.628520
14        14   0.912931  0.848745 -1.215276 -0.024202
15        15  -0.661489 -1.543994 -0.452754 -1.589521
16        16  -0.272063 -1.012084  0.414147 -0.021171
17        17   0.468146  0.232378 -1.890289  0.827622
18        18   1.744694  0.052282  0.890462 -0.155540
19        19   0.381029  0.666666  0.957832  1.476093
20        20   0.503970  0.567258  1.578919  1.041839
21        21  -0.561937  0.534243  0.246232  0.763021
22        22   0.064446 -0.483406 -0.584869 -1.066767

```

23	23	2.762885	0.060047	-0.709070	-1.350154
24	24	-0.399912	0.755714	-0.138098	0.771859
25	25	2.289132	-0.178023	0.392030	-1.638884
26	26	-1.667799	-0.723245	-0.298492	-0.464406
27	27	0.871473	-0.875910	1.693428	-0.120278
28	28	2.035892	-0.442804	1.710999	-0.164896
29	29	2.646331	0.006441	-0.819143	-0.140657
30	30	-1.645022	1.841754	-0.390258	-0.118842
31	31	-2.085251	0.043444	-0.183578	0.548298
32	32	1.427635	-0.349981	1.038814	-0.955178
33	33	-0.680210	0.566180	-0.325638	0.244383
34	34	-0.026502	0.758021	0.466427	-0.198146
35	35	0.433309	0.623019	-0.201789	1.983037
36	36	0.646673	-0.715421	-0.322057	0.645001
37	37	0.048681	-1.203145	0.460026	0.192644
38	38	0.098718	-0.672915	-0.229832	-0.187133
39	39	-0.059930	-1.472300	-1.281250	-0.666214
40	40	1.515724	0.207993	-1.037067	-0.393363
41	41	-0.949868	-1.115604	-0.554789	1.324991
42	42	-1.452316	-0.572414	-0.808790	1.615190
43	43	-1.117542	0.083952	-0.874761	0.884389
44	44	0.834118	0.423509	-0.565575	-0.169843
45	45	-0.382312	0.175994	-0.820539	-0.542783
46	46	-0.982393	-0.396464	0.550612	1.584936
47	47	0.581846	-0.994098	0.519327	1.286765
48	48	-0.412282	-1.889874	-0.047348	-0.767669
49	49	0.446292	0.326242	0.230500	0.841655

```
[33]: df.head(6)
```

```
[33]:
```

	Unnamed: 0	A	B	C	D
0	0	-1.557181	0.373416	1.040685	-0.648513
1	1	-0.835872	-0.005259	0.502099	-0.956839
2	2	-0.257153	2.638228	-0.535731	-1.069443
3	3	-0.505502	-0.339260	0.602622	-1.420995
4	4	1.366398	-0.368296	0.642091	1.609680
5	5	-1.350101	0.210097	-0.743280	0.035729

```
[35]: df.tail(5)
```

```
[35]:
```

	Unnamed: 0	A	B	C	D
45	45	-0.382312	0.175994	-0.820539	-0.542783
46	46	-0.982393	-0.396464	0.550612	1.584936
47	47	0.581846	-0.994098	0.519327	1.286765
48	48	-0.412282	-1.889874	-0.047348	-0.767669
49	49	0.446292	0.326242	0.230500	0.841655

```
[37]: df.shape
```

```
[37]: (50, 5)
```

```
[39]: df.mean()
```

```
[39]: Unnamed: 0    24.500000  
      A          0.020103  
      B         -0.123273  
      C         -0.044716  
      D          0.059140  
      dtype: float64
```

```
[41]: df.median()
```

```
[41]: Unnamed: 0    24.500000  
      A         -0.090116  
      B          0.000591  
      C         -0.215810  
      D         -0.091596  
      dtype: float64
```

```
[47]: df['A'].head(5).mode()
```

```
[47]: 0    -1.557181  
      1    -0.835872  
      2    -0.505502  
      3    -0.257153  
      4     1.366398  
      Name: A, dtype: float64
```